Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Applied business Analytics Practical

Practical – II

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Class: TYIT	Batch:1
Date of Assignment:31/07/2026	Date/Time of Submission:31/07/2026

2. Describing the Distribution of a Variable:

a. Obtain a dataset and calculate descriptive statistics (mean, median, mode, variance, etc.) for a specific variable of interest.

```
[58]
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[22]
df = pd.read_csv("Orders.csv")
```

```
[29]
print("Minimum :", shipping .min())
```

```
Minimum : 0.49
```

```
[30]
print("Maximum :", shipping .max())
```

```
Maximum : 164.73
```

```
[31]
print("Range :", shipping .max() - shipping.min())
```

```
Range : 164.23999999999998
```

```
[32]
print("Variance :", shipping.var())
```

```
Variance : 298.04749034418774
```

```
[33]
print("Standard Deviation :", shipping .std())
```

```
Standard Deviation : 17.264051967721475
```

```
[34]
print("Count :", shipping.count())
```

```
Count : 8399
```

```
print("Sum :", shipping .sum())
```

[35] 0.0s Python

```
Sum : 107831.04000000001
```

```
print("\nDescriptive Statistics")
print(shipping .describe())
```

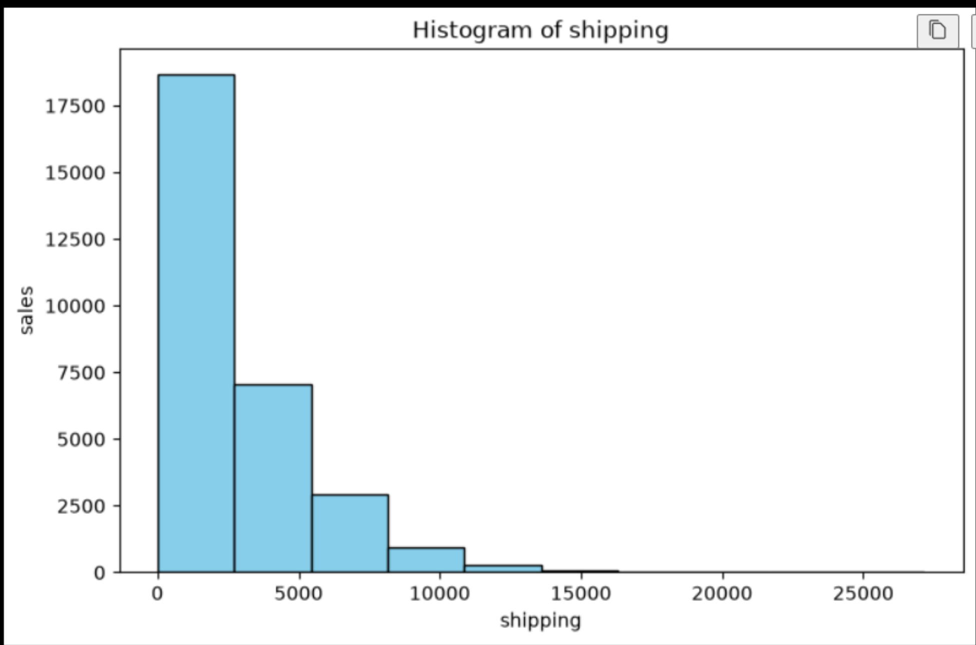
[36] 0.0s Python

```
Descriptive Statistics
count    8399.000000
mean       12.838557
std        17.264052
min         0.490000
25%         3.300000
50%         6.070000
75%        13.990000
max       164.730000
Name: Shipping Cost, dtype: float64
```

b. Create visualizations (histograms, box plots) to depict the distribution of a variable and analyze its characteristics.

```
plt.figure(figsize=(8,5))
plt.hist(sales, bins=10, color="skyblue", edgecolor="black")
plt.title("Histogram of shipping")
plt.xlabel("shipping")
plt.ylabel("sales")
plt.show()
```

[49] 0.3s Python



```python
import seaborn as sns
plt.figure(figsize=(8,3))
sns.boxplot(x=sales, color="orange")
plt.title("Box Plot of Sales")
plt.xlabel("sales")
plt.show()
```

[57] 5.1s Python